2025-2026 Spring Semester BIL216 Final Project

Phase 1 Technical Report: Sound Signal Analysis and Emotion Classification

This report summarizes the first working version of the proposed system.

Course	BIL216 Signals and Systems
Project	Sound Signal Analysis and Emotion Classification
Phase	Phase 1 - Beginning Model
Group No	18
Team Members	Mustafa Kemal Demiralay (220611032), Efe Dağ (220611015)
GitHub Link	https://github.com/mkdmrly/bil216-emotion-classification
Total Recordings	TBD
Initial Accuracy	TBD

1. Introduction

The aim of this project is to automatically classify five different emotions from speech recordings: neutral, happy, angry, sad, and surprised. In Phase 1, the goal is to build an initial working model and establish a signal-processing and machine-learning pipeline that can be improved in later stages of the project.

2. Dataset Characterization

Since the dataset has not yet been added to the project workspace, the class distribution could not be inserted automatically into this report. This section will be updated once the recordings are placed in the dataset folder.

This initial version is designed to process the recordings collected during the midterm project together with their emotion labels. The system supports both a folder-based dataset structure and a CSV-based labeling structure.

3. Methodology

The system was developed in Python. In the first stage, speech recordings are loaded as single-channel signals at a fixed sampling rate, and a set of acoustic features is extracted. The selected features include zero-crossing rate, RMS energy, pitch statistics, spectral centroid, spectral bandwidth, spectral rolloff, spectral flatness, spectral contrast, MFCC coefficients, and delta MFCC coefficients.

In the classification stage, several classical machine-learning models are compared. Support Vector Machines (SVM), Random Forest, K-Nearest Neighbors, and Logistic Regression are evaluated by cross-validation, and the best-performing configuration is selected. This approach provides a fast, interpretable, and extensible baseline for Phase 1.

4. Statistical Findings

After feature extraction, the system can generate selected feature distribution plots and a confusion matrix to observe separability between emotional classes. If the training pipeline has been executed, these visual outputs are automatically saved in the `results/` folder.

5. Classification Performance

Since the real dataset has not yet been placed in this workspace, numerical performance values could not be generated automatically at this stage. However, the baseline pipeline is ready, and once the dataset is added, the accuracy, classification report, and confusion matrix can be produced with a single command.

6. Error Analysis and Discussion

In speech-based emotion recognition, some emotions may be acoustically similar. In particular, neutral and sad may be confused because of lower energy levels and limited pitch variation. Although angry and happy can often be separated through increased energy, speaker differences and recording quality may still affect the results. In later phases, the system can be improved through feature selection, better class balance, and speaker-independent evaluation.

7. Classification Report Summary

Not generated yet.

8. GitHub Link

https://github.com/mkdmrly/bil216-emotion-classification

9. Resources Used

- Python
- librosa
- scikit-learn
- matplotlib
- GitHub
- Course notes
- Project brief

10. AI Prompts

- Prepare a Phase 1 starter project for speech emotion classification using classical machine learning methods.
- Create a short technical report draft in English for the BIL216 final project.

11. Team Members

- Mustafa Kemal Demiralay - Student No: 220611032
- Efe Dağ - Student No: 220611015

12. Team Member Contributions

- Mustafa Kemal Demiralay: Data collection, initial project structure, and report writing.
- Efe Dağ: Feature extraction, model comparison, and interpretation of the results.